

# MOHAMED BASSAM

AI Engineer | Machine Learning | Computer Vision | NLP

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🌐 Portfolio | 📁 GitHub

## PROFESSIONAL SUMMARY

**AI Engineer** with hands-on experience designing, developing, and deploying end-to-end AI applications across Machine Learning, Deep Learning, Computer Vision, and NLP. Proficient in Python, TensorFlow, PyTorch, YOLO11, and Flask, with a strong focus on building production-ready solutions—from data preprocessing and model training to deployment via REST APIs and interactive web dashboards. Dedicated to engineering intelligent, scalable systems that solve complex real-world problems.

## TECHNICAL SKILLS

### Programming & DB:

Python SQL SQLite JSON

### Deep Learning & NLP:

TensorFlow PyTorch Keras LSTM BERT Hugging Face

### Computer Vision & ML:

YOLO11 OpenCV Scikit-learn Regression/Classification

### Backend & Deployment:

Flask REST APIs Docker PythonAnywhere n8n (Webhooks)

### Tools & Frameworks:

Pandas NumPy CUDA Git & GitHub

## SELECTED AI PROJECTS

### AI Stock Price Prediction - AI Stock Market Analysis & Paper Trading Platform

Flagship Project

Tech: Python, Flask, TensorFlow/Keras, LSTM, Pandas, NumPy, Scikit-learn, Chart.js, REST APIs, Alpaca Paper Trading API

- Built an end-to-end AI stock analysis and paper-trading platform using LSTM forecasting, technical indicators, confidence scoring, and Flask dashboard deployment.
- Developed an explainable dashboard showing forecasts, BUY/SELL/HOLD signals, entry price, stop-loss, take-profit, risk level, expected move, and model reasoning.
- Integrated RSI, EMA, SMA, MACD, ROC, Bollinger Bands, ATR, VWAP, sentiment logic, and volatility analysis to support market decision-making.
- Added historical backtesting and walk-forward validation using return, win rate, drawdown, Sharpe ratio, directional accuracy, and prediction error metrics.
- Integrated Alpaca Paper Trading API with simulated order execution, portfolio tracking, trade history, trade journal, duplicate-order prevention, and bracket-order protection using take-profit/stop-loss exits.

## AI Industrial Safety Monitoring System

Computer Vision

Tech: Python, YOLO11, OpenCV, n8n, Webhooks, CUDA

- Engineered a real-time AI-powered industrial safety monitoring system utilizing YOLO11 and OpenCV.
- Implemented automated PPE detection workflows for Hardhat/No-Hardhat verification in workplace environments.
- Programmed intelligent Safe Zone and Danger Zone spatial monitoring loops featuring automated violation logs.
- Integrated dual-model validation logic to maximize detection reliability and minimize false alerts.
- Connected n8n cloud automation workflows to trigger instant Gmail safety alerts with automated incident image attachments.
- Optimized system inference speed using Torch CUDA GPU acceleration.

## AI Study Assistant – NLP Web Application

NLP

Tech: Python, Flask, Hugging Face Transformers (BART)

- Developed an NLP-based text summarization application utilizing pre-trained Hugging Face Transformers.
- Built a Flask backend with custom API endpoints capable of processing direct text inputs and multi-page PDF documents.
- Designed and coded text chunking algorithms to cleanly process large documents without encountering model memory constraints.

## AI Writer Web Application

Full-Stack AI

Tech: Python, Flask, SQLite, PythonAnywhere

- Developed and deployed a full-stack AI SaaS application for text generation, content rewriting, and user-driven text assistance.
- Implemented secure user authentication, session management, and persistent history tracking using Flask and SQLite.
- Developed RESTful API endpoints for handling core text generation logic and asynchronous user interactions.
- Deployed the live production application successfully on PythonAnywhere.

## PROFESSIONAL EXPERIENCE

### Machine Learning Intern

2024

CodeAlpha

- Developed, tested, and evaluated machine learning models using Support Vector Machines (SVM) and neural networks.
- Applied TensorFlow and OpenCV frameworks to build practical solutions for object recognition and data classification tasks.
- Evaluated model classification performance metrics utilizing ROC curves, confusion matrices, and precision-recall scores.
- Utilized NumPy, Matplotlib, Pandas, and Keras to manipulate large datasets and visualize experimental results.

## EDUCATION

### Bachelor of Computer Science (Artificial Intelligence)

Graduated: July 2025

Arab Open University (AOU) & The Open University (UK)

- Dual Degree Program verified by The Open University, UK.